

Sprint Society — Runner Classification System (Research Report)

Executive Summary

The current system is too simple: 3 buckets (beginner/intermediate/advanced) based solely on estimated VO2max. This doesn't reflect real running progression, which is a multi-year journey with measurable checkpoints.

Proposed system: 4 Tiers x 10 Levels = 40-point scale

Tier	Levels	Who	Timeline to Next Tier
Beginner	1-10	Never run → can run 5K continuously	2-6 months
Intermediate	1-10	Regular runner → competitive club runner	6-24 months
Advanced	1-10	Club competitive → sub-elite performance	2-5 years
Pro	1-10	Sub-elite → international/world class	5-15+ years

Pro 10 = Kipchoge-level. Most humans will never reach it. That's the point — it's a realistic quest.

Classification Factors (Multi-Dimensional Score)

Unlike simple VO2max-only systems, Sprint Society uses a **composite score** from 6 measurable factors:

1. Performance (40% weight)

Actual race times or equivalent time-trial performances. Most objective measure.

Level	Male 5K	Female 5K	Male Marathon	Female Marathon
B1	Can't run 1km	Can't run 1km	N/A	N/A
B5	35:00	40:00	N/A	N/A
B10	28:00	32:00	5:00:00	5:30:00
I1	27:00	31:00	4:45:00	5:15:00
I5	23:00	27:00	4:00:00	4:30:00
I10	20:00	23:30	3:30:00	4:00:00
A1	19:30	23:00	3:20:00	3:50:00
A5	17:30	20:30	3:00:00	3:25:00
A10	15:30	18:00	2:40:00	3:00:00
P1	15:00	17:30	2:35:00	2:55:00
P5	14:00	16:30	2:25:00	2:45:00
P10	13:00	15:00	2:05:00	2:20:00

Based on Jack Daniels VDOT tables + World Athletics age-grading percentages

2. VO2max / VDOT (20% weight)

Estimated from race performance or direct testing.

Level	VO2max (Male)	VO2max (Female)	VDOT Equivalent
B1	<30	<25	<25
B5	30-35	25-30	25-30
B10	35-40	30-35	30-37
I1	40-42	35-37	37-40
I5	42-48	37-43	40-48
I10	48-52	43-47	48-52
A1	52-55	47-50	52-55
A5	55-60	50-55	55-60
A10	60-65	55-60	60-65
P1	65-68	60-63	65-68
P5	68-73	63-67	68-73
P10	75+	68+	75+

3. Training Volume (15% weight)

Weekly km consistently maintained. Volume = aerobic engine size.

Level	Weekly KM	Sessions/Week
B1-B3	0-10	1-2
B4-B7	10-20	2-3
B8-B10	20-30	3-4
I1-I5	30-50	4-5
I6-I10	50-70	5-6
A1-A5	70-100	6-7
A6-A10	100-130	7-10
P1-P5	130-160	10-12
P6-P10	160-220+	12-14 (doubles)

4. Consistency (15% weight)

How many weeks per year the runner trains without interruption.

Level	Weeks/Year Active	Training Age (Years)
B1-B5	<20	0-0.5
B6-B10	20-35	0.5-1
I1-I5	35-42	1-2

I6-I10	42-48	2-4
A1-A5	48-50	4-7
A6-A10	50-52	7-10
P1-P10	50-52 (year-round)	10+

5. Lactate Threshold (5% weight)

Percentage of VO2max sustainable at threshold. Improves with training age.

Level Range	LT as % of VO2max
Beginner	60-70%
Intermediate	70-80%
Advanced	80-87%
Pro	87-92%

6. Running Economy (5% weight)

Oxygen cost at a given speed. Mostly genetic + biomechanics + years of running.

Measured indirectly through: pace-to-HR ratio improvement over time.

Level Advancement: How People Move Up

The Formula

Level Score = (Performance × 0.40) + (VO2max × 0.20) + (Volume × 0.15) + (Consistency × 0.15) + (LT% × 0.05) + (Economy × 0.05)

Each factor scored 1-40 (matching the total scale), then weighted.

Advancement Triggers

A runner advances a level when:

1. **Sustained performance** — hit the pace/time threshold for 3+ consecutive weeks (not a one-off good day)
2. **Volume milestone** — consistently hitting the weekly km for that level for 4+ weeks
3. **No regression** — hasn't dropped below current level's threshold in 2 weeks

Realistic Timelines (Based on Training Science)

Progression	Typical Duration	What Changes
B1 → B5	4-8 weeks	Build habit. Run/walk → continuous running.
B5 → B10	8-16 weeks	Build base. Run 5K without stopping. Add frequency.
B10 → I1	2-4 months	First structured training. Add tempo runs.
I1 → I5	6-12 months	Introduce intervals. Race a 5K. Training cycles.
I5 → I10	12-24 months	Higher volume. Periodization. Multiple race distances.
I10 → A1	6-12 months	Serious training. 6+ days/week. Racing regularly.

A1 → A5	1-3 years	High volume. Altitude/heat training. Strength work.
A5 → A10	2-5 years	Peak training. Professional coaching. 100+ km/week.
A10 → P1	2-4 years	Sub-elite. National level. Training camps.
P1 → P5	3-5 years	Elite. International competition. Full-time runner.
P5 → P10	5-10+ years	World class. 99.99th percentile. Genetic + dedication.

Total journey B1 → P10: 15-25+ years of dedicated training. This is realistic — Olympic marathoners typically train 10-15 years before peaking.

Training Prescription by Level

Beginner (1-10): Build the Habit

Sub-level	Focus	Weekly Structure	Key Principle
B1-B3	Habit formation	2-3 run/walks, 15-30 min	"Just get out the door"
B4-B6	Continuous running	3 runs, 20-35 min all easy	"Run slow to run long"
B7-B9	Base building	3-4 runs, 30-45 min, 1 slightly longer	"Add 10% volume/week max"
B10	Race ready	4 runs, including first tempo effort	"You can now race a 5K"

Training split: 100% easy pace. NO intervals. NO speed work. ONLY aerobic development. **Coach personality at this level:** Encouraging, celebrating small wins, removing barriers.

Intermediate (1-10): Build the Engine

Sub-level	Focus	Weekly Structure	Key Principle
I1-I3	Introduce structure	4 runs: 3 easy + 1 tempo (20 min)	"Quality within quantity"
I4-I6	Speed introduction	4-5 runs: 3 easy + 1 tempo + 1 interval	"Your body adapts to stress"
I7-I9	Volume + intensity	5-6 runs: 3 easy + 1 long + 1 tempo + 1 interval	"Periodize in 4-week blocks"
I10	Race specialization	5-6 runs: structured by race distance	"Train specifically for your goal race"

Training split: 80% easy, 10% tempo, 10% intervals (polarized model). **Coach personality:** More technical. Explaining training zones. Introducing data.

Advanced (1-10): Optimize Everything

Sub-level	Focus	Weekly Structure	Key Principle
A1-A3	High volume base	6-7 runs, 70-90 km/week, doubles start	"Volume is the foundation of speed"
A4-A6	Lactate threshold focus	6-7 runs, threshold sessions 2x/week	"Push the ceiling higher"
A7-A9	Peak specific training	7+ runs, race-specific blocks, altitude	"Every session has a purpose"
A10	Championship prep	10+ sessions/week, tapering mastered	"Peak when it matters"

Training split: 80% easy, 8% tempo, 8% VO2max, 4% race pace. **Additional:** Strength training 2x/week, mobility, nutrition timing, sleep optimization. **Coach personality:** Analytical. Data-heavy. Marginal gains focus.

Pro (1-10): World-Class Performance

Sub-level	Focus	Weekly Structure	Key Principle
P1-P3	National level	130-150 km/week, periodized blocks	"Recovery is as important as training"
P4-P6	International level	150-180 km/week, altitude camps	"Race selection is strategy"
P7-P9	Sub-2:10 marathon	180-200+ km/week, full-time	"Every 1% matters"
P10	World record contender	200+ km/week, bespoke everything	"No human is limited"

Training split: Varies by coach/methodology. Typically 85% easy, 15% quality. **Additional:** Full-time athlete. Team of physios, nutritionists, biomechanists, coaches. **Coach personality:** Strategic partner. Race tactics. Peaking cycles. Competition intelligence.

What Makes This Different From Competitors

Feature	Strava	Nike Run Club	Garmin	Sprint Society
Classification	None (just segments)	4 levels (basic)	Training status only	40-point composite scale
Multi-factor	No	No	Partial (load + recovery)	6 factors weighted
Realistic timeline	No progression	No timeline	No	Months/years displayed
Sub-levels	No	No	No	10 within each tier
What to improve	Not told	Generic plans	Vague suggestions	Specific next-level criteria shown
Training by level	Same for all	Same 4-week plans	Basic zones	Level-specific prescription

How Sprint Society Shows This to Users

Level Card (Profile)

INTERMEDIATE • Level 6

6.3/10

Next: I7 (est. 6 weeks)

Need: 5K under 22:00 + 45km/week

VO2max: 46.2 | LT: 77%

Volume: 42km | Consistency: 89%

Advancement Panel (What's Needed)

To reach Intermediate 7:

✓ 5K pace under 4:30/km (done!)

○ Weekly volume 45km+ (you: 42km)

○ Sustained for 4 weeks (week 2/4)

○ No missed weeks (streak: 6)

Estimated: 6 weeks at current rate

Journey Map (Train tab)

Shows the full 40-level path. User sees where they are, where friends are, what's ahead. Creates aspiration without discouragement.

Age Grading

Since a 40-year-old and a 20-year-old shouldn't be compared raw:

- All performance benchmarks are **age-graded** using World Athletics tables
- A 35:00 5K for a 16-year-old = Beginner 5
- A 35:00 5K for a 55-year-old = Intermediate 3 (age-adjusted)
- This means EVERYONE can progress regardless of age

Implementation Changes Needed

Backend (ai-profiler.ts)

1. Replace 3-tier system with 40-level composite scorer
2. Add `calculateLevel(performance, vo2max, volume, consistency, lt, economy)` function
3. Store `runner_level` (1-40), `tier` (B/I/A/P), and `sub_level` (1-10)
4. Track progression history (level changes over time)
5. Calculate "next level" requirements for each user

Frontend (Train tab + Profile)

1. Level card showing current position on 40-point scale
2. "What's needed for next level" checklist
3. Estimated time to next level
4. Journey map visualization (all 40 levels as a path)

Training Engine

1. Training prescription varies by exact sub-level (not just tier)
2. Intensity distribution shifts as level increases
3. Volume recommendations scale with level
4. Recovery advice adapts to training age

Sources & Methodology

This framework is built from:

- **Jack Daniels' Running Formula** — VDOT tables, training pace calculation
- **Pfiztinger's Advanced Marathoning** — volume/intensity by level
- **World Athletics age-grading tables** — performance normalization
- **Norwegian method research** — polarized training distribution
- **Long-term athlete development (LTAD) models** — progression timelines
- **Billat et al.** — VO2max/vVO2max relationship to performance
- **Seiler (2010)** — intensity distribution in elite runners
- **IAAF scoring tables** — performance point equivalents

Questions for Ishan Before Implementation

1. Do we show the FULL 40-level scale upfront? Or reveal tiers gradually (gamification)?
2. Should "Pro" tier be aspirational only (shown but not achievable for most) or hidden until Advanced 8+?

3. Age-grading: apply automatically, or let users opt-in?
4. Should we factor in **race participation** (actual races, not just training runs)?
5. How aggressive should level-down be? (If someone stops training for a month, do they drop levels?)